

Chapter 5: Vector and Raster based Spatial Analysis

1. Introduction to Spatial Data Analysis

Spatial data analysis is the core subsystem of a GIS that distinguishes it from a simple computerized drafting (CAD) package or database management system. It is the process by which raw geographic data is transformed into useful information.

As highlighted by **Stan Aronoff**, spatial analysis extracts meaning from data by uncovering spatial patterns, evaluating distributions, identifying anomalies, and modeling real-world processes. It answers critical geographic questions such as:

- **Location:** *What is at this specific site?*
- **Condition:** *Where do specific environmental criteria exist together?*
- **Trends:** *What has changed in an area over a specified time period?*
- **Patterns:** *Are certain geographic anomalies clustered or dispersed?*
- **Modeling:** *What happens to the surrounding area if a new dam is built?*

2. GIS Usage in Spatial Analysis

GIS facilitates spatial analysis by automating geometric and statistical processes that are physically impossible or highly inefficient to perform manually on paper maps. It allows for:

- **Thematic Overlaying:** Stacking multiple independent layers aligned to the same coordinate reference system to evaluate complex interactions.
- **Objective Quantification:** Accurately calculating distances, perimeters, land areas, slope gradients, and network paths using absolute coordinate math.

- **Scenario Simulation (What-If Modeling):** Dynamically altering parameters (e.g., increasing a flood zone height) to visualize and assess prospective risks before taking physical action.

3. Vector-Based GIS Analysis

Vector analysis operates on discrete objects using exact coordinate math (X,Y). The analysis relies heavily on geometric calculations and the database links between a shape and its attribute row.

A. Buffering (Proximity Analysis)

Buffering calculates an area of exclusion or influence around a feature at a specified uniform distance.

- **Point Buffer Example:** * *Scenario:* Setting up a public health notification zone.
 - *Application:* A town discovers a contaminated public water well. The GIS analyst selects the well point and generates a **500-meter buffer zone** around it to identify and alert all household properties falling within that radius.
- **Line Buffer Example:** * *Scenario:* Environmental protection and construction zoning.
 - *Application:* To enforce a local environmental conservation law, a **100-meter buffer** is generated along a major river network centerline. The resulting polygon represents a protected riparian zone where new construction is legally prohibited.
- **Polygon Buffer Example:** * *Scenario:* Urban noise control mapping.
 - *Application:* Generating a **1-kilometer buffer zone** around an airport runway boundary polygon to delineate areas where residential housing developers must install specific acoustic soundproofing materials.

B. Vector Overlay Operations

Overlay analysis stacks independent thematic layers that share the exact same coordinate system to create an entirely new vector dataset containing combined spatial shapes and attributes.

- **CLIP (The "Cookie Cutter"):**
 - *How it works:* Extracts input features that fall strictly within the boundary polygon of a clip layer. It changes the geometry of the input layer but **does not** combine attribute data.
 - *Example:* You have a massive national dataset containing all roads in Ethiopia (Input), and a polygon layer of your specific administrative District (Clip). Running a Clip isolates and extracts *only* the roads that lie inside your district, giving you a clean, manageable local road layer.
- **INTERSECT:**
 - *How it works:* Preserves only the overlapping geometric portions common to *both* input layers. The output attribute table is combined, containing data from both inputs.
 - *Example:* You overlay a **Forest Reserve Polygon Layer** with a **Wildfire Burn Extent Polygon Layer**. The Intersect operation outputs *only* the areas where the forest reserve and the fire overlapped, allowing you to calculate the precise acreage of protected timber lost, alongside the tree species attributes.
- **UNION:**
 - *How it works:* Combines all geometric features and all attribute data from both input polygon layers, whether they overlap or not.
 - *Example:* An urban planner overlays a **1996 Land Use Layer** with a **2026 Land Use Layer** using Union. The output map contains all boundaries from both years. Where they overlap, the attribute table shows both the old land use class and the new land use class side-by-side, allowing the planner to systematically track 30 years of urban expansion.
- **ERASE:**
 - *How it works:* Removes input features that fall inside the boundary of an erasing polygon layer.
 - *Example:* You want to find suitable areas to build a school. You take a **Total Available Land Layer** (Input) and run an Erase operation using a **Swamp/Wetland Polygon Layer** (Erase). The system deletes all swampy areas from your asset pool, leaving you with dry land options.

2. Raster-Based GIS Analysis

Raster analysis operates on a continuous matrix grid of cells (pixels). Because cells are uniform and co-registered (perfectly aligned on top of each other), the operations treat layers like mathematical matrices.

A. Map Algebra

Map algebra executes mathematical, statistical, or logical functions on one or more raster grids cell-by-cell.

- **Arithmetic Grid Operation Example:**

- *Scenario:* Measuring structural deformation or volcanic inflation/deflation.
- *Application:* A geomatics engineer takes a Digital Elevation Model (DEM) grid from January 2025 (Grid A) and subtracts it from a new DEM grid captured in June 2026 (Grid B).

Output Grid=Grid B–Grid A

A positive cell value in the output grid reveals a physical rise in the ground surface (inflation), while a negative value reveals land subsidence or erosion.

- **Logical/Boolean Grid Operation Example:**

- *Scenario:* Land suitability site selection for an agricultural development hub.
- *Application:* You have a **Slope Gradient Grid** (where cells hold % slope values) and a **Rainfall Grid** (where cells hold annual mm values). You execute a logical query:

Output Suitability = (Slope $\leq$ 5) AND (Rainfall $\geq$ 1200)

The computer evaluates every pixel coordinate. If a cell matches both criteria, it outputs a value of 1 (Suitable); if not, it receives a 0 (Unsuitable).

B. The Four Analytical Spatial Classes

- **1. Local Operations (Cell-by-Cell):**

- *Mechanism:* The output value of a single cell depends *only* on the input value(s) at that exact same coordinate position.
- *Example:* Converting an elevation raster layer from feet to meters. The system processes each individual cell independently using a constant multiplier:

$$\text{Output Cell} = \text{Input Cell} \times 0.3048$$

- **2. Focal Operations (Neighborhood):**

- *Mechanism:* The output value of a target cell is computed using its own value plus the values of its immediate surrounding neighbors (typically using a moving 3×3 or 5×5 pixel filter window).
- *Example: Slope and Aspect Derivation.* To calculate the slope of a central cell, the software analyzes the elevation differences among its 8 surrounding neighbor cells to determine the maximum rate of change in elevation.
- *Another Example: Image Smoothing.* Replacing a central cell's value with the *mean average* value of its 3×3 neighborhood to eliminate random sensory background noise from satellite imagery.

- **3. Zonal Operations:**

- *Mechanism:* Computes a spatial statistic (mean, max, sum) for an input data grid across distinct geographic zones defined by an entirely separate, thematic boundary layer (vector or raster).
- *Example:* Evaluating agricultural productivity across administrative boundaries. You take a continuous **NDVI Vegetation Index Raster Grid** and a vector **District Boundary Layer**. The Zonal Statistics tool groups all the pixels falling inside each district polygon and calculates the *Mean NDVI Score* per district, allowing regional managers to see which district is experiencing the worst crop drought.

- **4. Global Operations:**

- *Mechanism:* The output value of every single cell is mathematically dependent on the data characteristics of the *entire* input grid combined.
- *Example: Euclidean Distance Modeling.* You have a single cell representing a newly proposed landfill site. A global distance operation scans the entire map canvas to calculate the exact straight-line distance from *every single pixel* on the map back to that landfill location, generating a continuous distance gradient across the region.